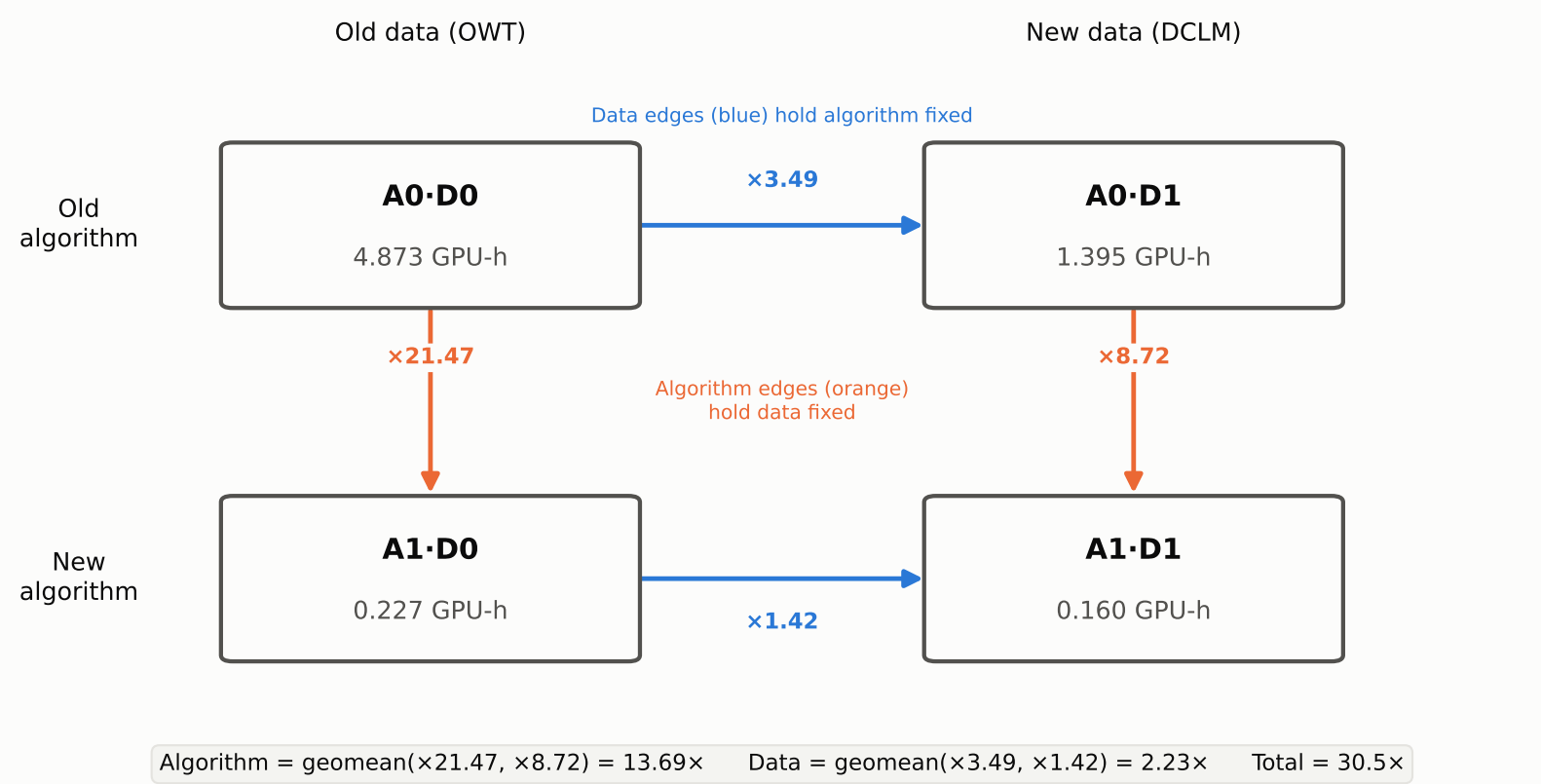


The 2x2 factorial and its log-space Shapley split



Each edge is a GPU-hours ratio between two arms that differ in a single intervention; the Shapley multiplier for an intervention is the geometric mean of its two edges (both orderings averaged). Worked example at the 124M GPT-2 baseline scale, current training recipe. Values re-derived from the four arm GPU-hours in results/small/ceg_newdef.json (held-out Wikipedia evaluation set, "wiki_eval").